

ResiliTrack AI

Anticipatory Disaster Intelligence Platform

Track 5: Cyclone Impact & Infrastructure Vulnerability Forecaster

Google Earth Engine | Gemini 1.5 Flash | OpenStreetMap | Streamlit Cloud

Build with AI: Code for Communities - 2nd Edition

THE PROBLEM

Why generic cyclone warnings fail coastal communities

01 Blind Spot in Last-Mile Intelligence

Weather cones cover hundreds of sq km but cannot tell if a specific hospital feeder road will be under 1.5 m of water at 3 AM — the critical hour before landfall.

02 Cascading Infrastructure Failure

Disaster commanders cannot identify which emergency shelters will be cut off before flooding occurs, making it impossible to pre-position life-saving resources.

03 Communication Bottleneck

Emergency officers receive raw satellite imagery without AI synthesis into immediate, localized, multi-lingual tactical actions for frontline workers and citizens.

THE SOLUTION - ResiliTrack AI

Shifting from reactive recovery to anticipatory pre-landfall action

SIMULATE

Storm Surge & Hydro-Elevation Mapping

GEE NASADEM/SRTM elevation matrices combined with wind speed & 24h rainfall accumulation to generate hyper-local flood depth overlays.

ANALYZE

Infrastructure Exposure Graph

Extracts hospitals, power grids, bridges from OpenStreetMap. Calculates accessibility decay and flags isolated trauma centers.

DECIDE

Gemini AI Operations Intel

Synthesizes satellite maps + telemetry into executive reports, priority evacuation zones, and multi-lingual citizen alerts.

FUND

Parametric Insurance Score

Pre-landfall severity index triggers instant emergency liquidity payouts BEFORE the storm makes landfall.

THE GEMINI AI ADVANTAGE

Gemini 1.5 Flash - context-aware, proportional, multi-lingual Virtual Disaster Commander

What Gemini Produces:

- > Structured executive risk reports
- > Priority evacuation zone commands
- > Infrastructure isolation assessments
- > Multi-lingual citizen SMS dispatches
Bengali / Odia / Hindi / English
- > Parametric insurance severity index

Edge-Case Handling (Refined Prompts):

Minor Storm (<119 km/h)

Scales to MODERATE or LOW. Focuses on preparedness, avoids panic.

Zero Infrastructure Risk

Confirms critical infrastructure is secure and fully operational.

Below Catastrophe Threshold

Sets payout to NOT MET with \$0 — no false triggers, ever.

API Key Absent

Gracefully falls back to pre-built realistic simulation engine.

COMMUNITY IMPACT

Real-world value for millions of vulnerable coastal residents across South & Southeast Asia

Saves Lives

Hyper-localized evacuation orders — not generic warnings. Residents in low-lying fishing hamlets and riverine delta islands get specific, actionable guidance hours before landfall.

Protects Critical Infrastructure

Pre-landfall alerts ensure backup generators are deployed and amphibious transport is pre-positioned for isolated trauma centers and hospitals before flooding occurs.

Inclusive Communication

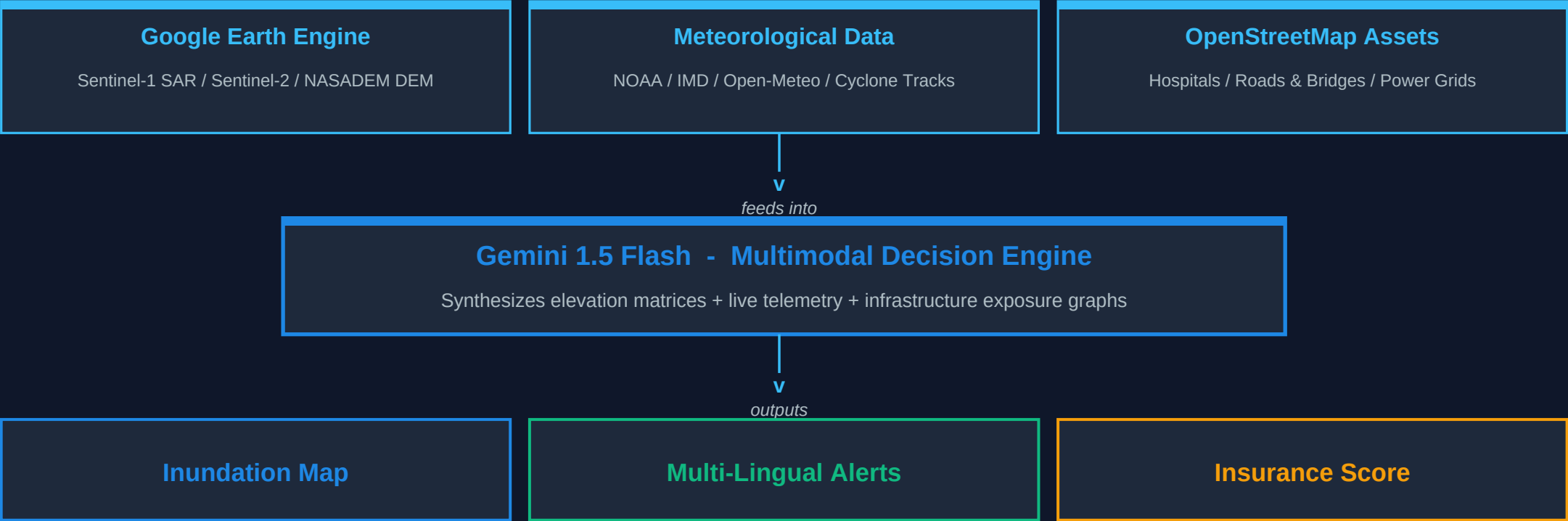
Auto-generates native-language alerts in Bengali, Odia, Hindi, and English — reaching the most vulnerable communities who may not understand broadcast warnings.

Financial Resilience

Parametric insurance scores trigger immediate liquidity payouts BEFORE the storm hits — communities buy supplies instead of waiting weeks for post-disaster aid approval.

SYSTEM ARCHITECTURE

End-to-end data flow: Satellite + Telemetry -> Gemini AI -> Operational Outputs



LIVE DEMO WALKTHROUGH

Live app: <https://efomw9k5jde5gwkxe3ybu.j.streamlit.app/>

01

Select Target Coastal Sector

Use the sidebar dropdown to pick a region (e.g. Bay of Bengal - Odisha/West Bengal). The interactive GIS map re-centres instantly.

02

Adjust Cyclone Severity Parameters

Move sliders: Max Wind Speed, Storm Surge Height, 24h Rainfall. Watch the flood polygon intensity and infrastructure markers update dynamically.

03

Run Gemini Multimodal Risk Synthesis

Click 'Run Gemini Risk Synthesis'. Gemini generates a full executive report, evacuation zones, tactical commands, and dispatches in 4 languages - Bengali, Odia, Hindi, English.

04

Explore Analytics & Submission Tab

Review Asset Inundation Depth charts, the Detailed Infrastructure Exposure Table, and the built-in hackathon submission description ready to copy-paste.

ROADMAP & CLOSING

Near-Term

Integrate live IoT river gauge sensors for real-time ground truth validation of flood depth predictions alongside GEE satellite feeds.

Mid-Term

Build automated WhatsApp / Telegram bot for direct citizen dispatch, integrating with national emergency alert systems (IMD, NDMA).

Long-Term

Expand full APAC coverage: Philippines, Vietnam, Bangladesh, Myanmar — becoming the region's go-to anticipatory disaster intelligence layer.

"With ResiliTrack AI, we aren't just weathering the storm - we are outsmarting it."

GitHub: github.com/swatiicfai/ResiliTrack-AI-CycloneShield

Live App: <https://efomw9k5jde5gwkxe3ybuj.streamlit.app/>